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**How can a large Dutch energy supplier predict which
 customers are most likely to churn?**

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1. Managerial Summary

The objective of this report is to determine what factors play a role in establishing what leads to an increasing churning rate for energy suppliers. Accordingly, it has been hypothesized that the variables age, income, electricity and gas usage, start-channel, and home-label have a significant effect on churn. Preliminary analysis concludes that these variables do indeed have a significant relationship with churn, meaning that they can be included in the more extensive mathematical models that are developed to predict churn. More specifically, age (age-group) and home-label have a negative relationship to churn while income and electricity and gas usage have a positive relationship, and people that register through the start-channel: website are more likely to churn than those who registered over the phone. Following this, a baseline model acting as a basis from which five other models have been developed. These models are a logistic regression model, a stepwise regression model, a tree model, a random forest model, and a SVM model. Subsequently, it can be concluded that for managers the tree model is recommended to use.

2. Define the Business Problem

2.1 Business problem

A large Dutch energy supplier wants to predict which customers are most likely to churn. The defining problems are that customers are churning at a rate approximately of 50% and the company does not have the appropriate model to predict churn.

3. Design the research

3.1 Management question

To define the management question (Blumberg et al., 2014), Dutch energy supplier's dilemma has been restated in a question form that is action oriented. Its main concern is to predict which customers are most likely to churn. Thus, the following management question has been formulated:

How does the Dutch energy supplier predict the likelihood of churning to potentially decrease the churn rate going forward?

3.2 Research question

According to Blumberg et al. (2014), the research question should be information-oriented and aims to outline what research should be conducted. Since the business challenges (Kooge et al. 2021) are frequently linked to the various stages of the customer lifecycle, it is crucial to include internal factors that affect the churn, such as the start channel through which the person became a customer. Taking into consideration the internal factors the following research question has been formulated:

To what extent is the churn influenced by internal factors?

3.3 Sub-questions

Sub-questions should be created in order to deconstruct the management question. Therefore, Kooge et al. (2021) recommends using the five W-questions (Who, What, When, Where and Why) that could be used to break down the business problem. The most important variables that can be found in the data for each sub-question, should be developed after the sub-question has been defined. For this paper three to five factors that literature shows to be most important are selected to include for further statistical analysis. For each selected factor, literature-based hypotheses are established to function as a guide for the analytical process.

Literature of the variables:

Age: The ‘who’ will be discussed. For this paper, the question can be asked; Who has a high churn rate? Subsequently, two variables seem to be highly relevant to consider: Age and Income. When it comes to age, a previously performed case study based on the UK market shows that younger people are more likely to switch energy suppliers or renegotiate their tariff (Citizens Advice, 2017). This research shows there is a 10% difference between 75+ aged people switching and all other age groups (Citizens Advice, 2017, p7). This suggests that this leads to the hypothesis: Age has a negative relationship with churn.

It has to be noted that relationship length is considered an interesting variable in the context of this case. However, since the nature of the variable relationship length is strongly connected with the variable age (someone that is young cannot have a long relationship length while someone older is more likely to have a longer relationship length)suggesting multicollinearity issues, it has been concluded that the variable relationship length will not be included.

Income: Literature seems to be inconsistent in their conclusions of whether high or lower income influences the likelihood of someone being more likely to switch energy suppliers (Deller et al., 2021). Research in many behavioral fields has shown that income is an important determinant in switching decisions (Deller et al., 2021). When the research is specifically focused on the UK, income does show a significant effect on the likelihood of someone switching energy suppliers or renegotiating their tariff. Namely people that earn £35k or over (highest category) have a 35% chance of engaging in such behavior, meanwhile the lowest category (below £16k) show the lowest likelihood (approximately 21%)(Citizens Advice, 2017). This leads to the hypothesis: Income has a positive relationship with churn.

Electricity and gas usage: This variable is a combination of the electricity usage and gas usage.

The two variables have their outliers dealt with and are then standardized so that they are on a similar distribution. The standardized values are then combined by finding the average value between the two for each observation.

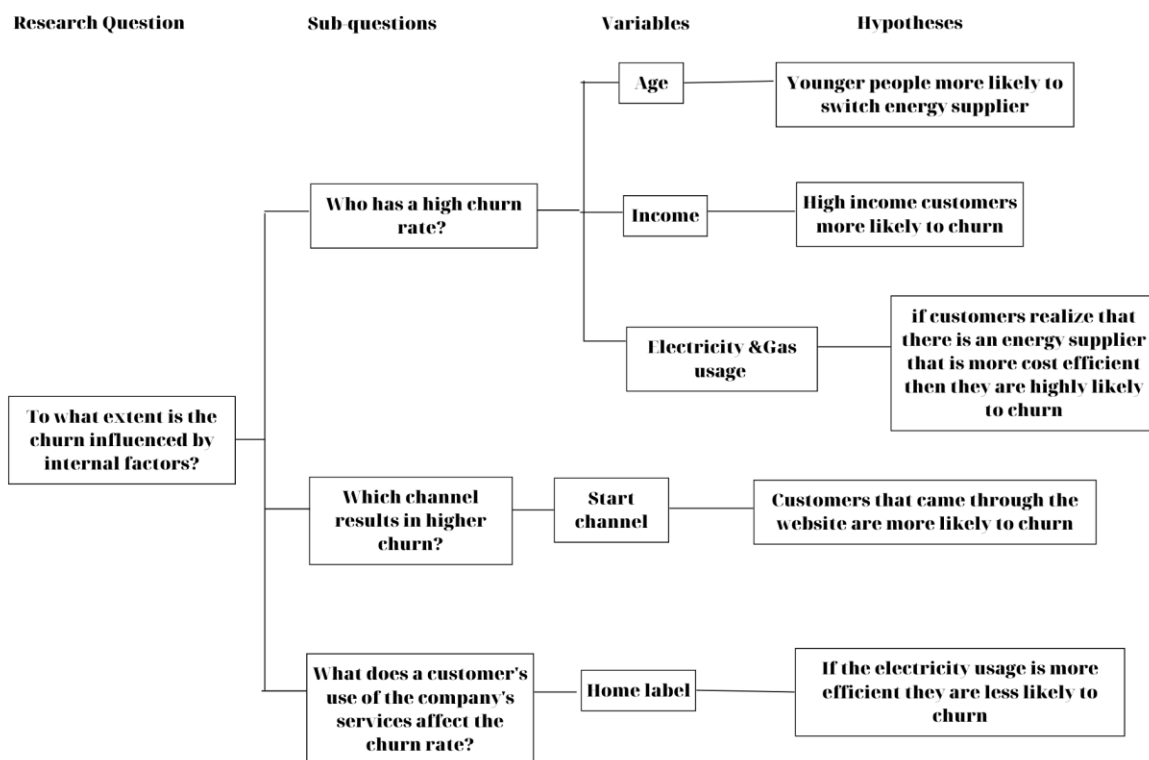
Consumers that have a high usage can proportionally save more in costs when they switch to an energy supplier that has lower tariffs. Hence, these people are considered to be more price sensitive and aware of their monthly usage/spending. Accordingly, if they realize that there is an energy supplier that is more cost efficient then they are highly likely to churn. For that reason, the following hypothesis has been created: Electricity and gas usage has a positive relationship with churn.

Start Channel: ‘What communication channel results in higher churn rates?’ This allows for exploration of the start channel variable. Exploration of customer behavior in the Netherlands has shown that the vast

majority of the people that switch energy suppliers do so through the website and not through customer service channels such as phone, at the door, or in stores (Nugteren, 2020). This leads to the hypothesis that customers that registered through the website are more likely to churn.

Home label: Next, the question ‘How do home characteristics of the customer affect the churn rate?’ can be formulated. There are two relevant home characteristics that can be explored: home label or home age. As these variables are theoretically too correlated only one will be chosen to use in the further analysis, Home label. Literature does not seem to show whether home labels have a direct effect on the likelihood of someone churning. However, it is argued that having a better home label (A is the best and G the worst) is more price efficient (Ruijter, 2022). This could indicate that people that have better home labels are less price sensitive and are therefore less likely to churn. Restating this in a hypothesis form gives: Home label has a negative relationship with churn. Additionally, research shows that having a certain label is an important determinant in the decision-making process for customers, hence it could be further explored how the variable home label is positioned with regards to the other variables.

Figure 1: Opportunity tree



4. Data preparation

4.1 Cleaning the data

Cleaning and transformation of the data

For cleaning the data, transformation of the variables into its correct format has been done. Most of the variables were already provided in the correct form, so changes occurred in the variables of Age, Income, Star Channel and Home Label.

Age: divided the age into four separate age groups, group 1 of age till 30, group 2 of age 31 till 50, group 3 of age 51 to 66 and finally group 4 of age above 67. These increments are chosen according to the life stages that someone is in. Until 30 can be seen as youngsters that are early on in their careers, a high percentage of this group is likely to still rent a house or switch houses frequently. Then, 31 till 50 is the category of middle-aged people that are in the midst of their life, most likely have bought their own house that they will live in for a longer period of time. Then, 51 to 66 are the older generation that have (most likely) lived in their house for a long time, have experience with energy labels and usage, and are most likely to be the highest earners. Lastly, the line is drawn at 67 because this is the legal age that people can retire in the Netherlands, hence everyone of this age and above can be categorized as retired.

Income: divided the income group according to the first and third quartile. Lowest class is below the first quartile, then 'lower middle class' is from the first quartile until the median, then 'upper middle class' is the median until the third quartile, and finally the 'highest class' is the third quartile and above. The classes are given in levels so that the lower class is given a value 1, the upper middle class is given a value 2, etc. So, the variable has four levels.

Electricity and gas usage: This variable is a combination of the electricity usage and gas usage. To create this variable the first step was to standardize the individual variables; electricity usage and gas usage. This is done because these two variables are in different metrics. Any aggregation of the two without this step would underestimate the effect of gas and overestimate the effect of electricity. After this, for each row, the mean of the two standardized variables is taken to create the new variable that is the combination of electricity usage and gas usage.

Start Channel: first, creating dummy variables by turning all the "Online" into 1 and "Phone" into 0. Afterwards, Star Channel was transformed into numeric.

Home Label: knowing that home label is ranged from 'A' (good) to 'G' (bad), creating variable dummy by turning 'A' into 1, 'B' into 2, 'C' into 3, 'D' into 4, 'E' into 5, 'F' into 6 and 'G' into 7 would make easier the further analysis. Afterwards, Home Label was transformed into numeric.

4.2 Dealing with missing and outliers

Dealing with missings

There is no missing data so dealing with the outliers is the next step.

The variables that are included are age, income, electricity and gas usage, start channel, and home label. It is unnecessary to look at outliers for age as age group will be used instead of doing a linear regression. Same goes for income as income group will be used. As for start channel it is a dummy which cannot possibly have outliers. Lastly home label is a categorical variable displayed in levels. The only variable that needs to be dealt with is the electricity and gas usage.

Using the descriptive statistics, the outliers are detected. The difference between the third quartile and the first quartile is called IQR. If a value over the third quartile + 1.5*IQR or below the first quartile - 1.5*IQR then it is an outlier. Those observations that were found to be outliers were then dealt with by using truncation. They were given the highest or lowest value which is not considered an outlier. If they were outliers on the higher end of the distribution, then they were given the highest value and vice versa. The lower limit for electricity usage was 792 kWh and the upper limit 3760 kWh. The lower limit for gas usage was -218 cubic meters and the upper limit 2618 cubic meters.

5. Explorative analysis

5.1 Correlation plot

Firstly, looking at the correlations of the variables gives positive feedback to our hypotheses. Churn seems to have a small to moderate negative relationship with contract length, income, income group, age and relationship length while having a positive relationship with electricity and gas usage.

5.2 Age

The variable Age is a nominal-continuous variable. Conducting the summary statistics of age, it is observed that the lowest age of customer is 17 years old, while the highest is 104. The histogram of age looks roughly the same on either side of its center, the mean and the median are approximately the same. As a result, the data of age is symmetrically skewed. Subsequently, a simple linear regression can be conducted to determine the correlation between age and churn. The results of the test conclude that age is significant because the p-value is less than 0.05 ($p < 2e-16$) with a coefficient of -0.0031451. This indicates that the relationship is negative, meaning that when age increases with one unit, churn will decrease by 0.003145. This is consistent with the hypothesis stated in section 2.

Figure 2: Correlation plot



5.3 Age group

The categories for this variable and the amount of observations in each category are: 30 and below (1341 observations), 31 till 50 (11409), 51 till 67 (5464), and 67 and above (1786). A logistic regression shows that the relationship between age group and churn is significant and negative (coefficient = -0.18170, $p < 2.e-$

16<0.05) To be more specific, 30 and below has the highest average churn being approximately 57%, 31 till 50 has an average churn of approximately 50%, 51 till 66 has an average churn of approximately 47%, and 67 and above has the lowest average churn being approximately 40%. This supports the hypothesis that age has a negative relationship with churn.

5.4 Income

The income variable is a numerical variable. Income is measured as euros per month. The minimum income is 684 euros a month. The first quartile of the Income is 3146 euros a month indicating that the lowest 25% earners are below 3146 euros a month. The highest 25% earners are earning over 6302 euros a month. The average income is 4987 euros per month. Since truncation was used on income to deal with outliers the maximum income is now 11034 euros but before that we had a maximum income of 235350 euros per month. The income variable has a moderate positive skew indicating that a bigger portion of our customers are towards the lower end of our distribution. Income has a significant relationship with churn, the p-value being $-1.85e-08 < 0.05$. Indicating that the higher the income the less likely the customer is to churn. This is not in line with the earlier stated hypothesis that stated income has a positive relationship to churn. Income has neither a significant relationship with electricity usage nor gas usage. The age has a significant effect on income ($p=2e-16 < 0.05$) with each year providing on average a 12 euro per month increase. Income has a significant effect on relationship length ($p=1.36e-07 < 0.05$). It should be mentioned that these two variables do not have a significant relationship prior to the truncation of income.

5.5 Income group

The income variable is a categorical variable with four levels. Each income group has around five thousand customers, so a sampling bias is not present. To assess whether an income group has a significant relationship with churn a logistic regression has been used. The results show that income group has a positive significant relationship to churn (coefficient = 0.01286, $p < 2e-16$). More specifically, lower class has an average churn of approximately 58%, lower middle class has 52%, upper middle class has 45%, and the upper class has 40%. This shows that the lower the class the higher the churn, indicating a negative relationship, which does not support the initial hypothesis that expects a positive relationship.

5.6 Electricity and gas usage

Due to the variable containing standardized values it does not add significant value to discuss the descriptive statistics. A logistic regression is used to evaluate its relationship to churn. This test shows that electricity and gas usage has a positive significant relationship to churn ($p < 2e-16$). More specifically, when electricity usage and gas usage increases by one unit, churn will increase by 0.98483. This supports the hypothesis stated in the previous section.

5.7 Start channel

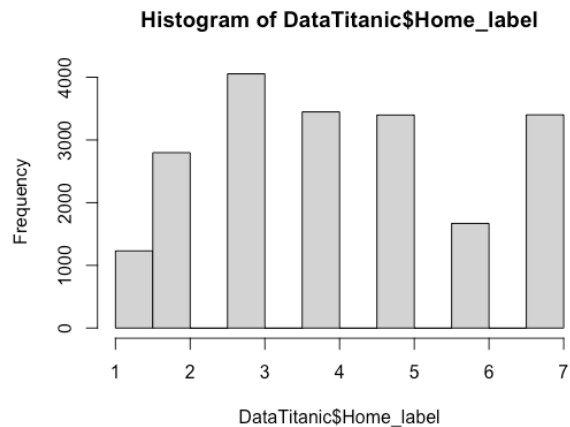
The variable start channel can either take on the value 'Online' or 'Phone' indicating if the contract was filled out by the customer on the firm's website (Online) or by calling up the firm (Phone). When looking at the distribution amongst the customers, it shows that approximately 82% filled out the contract online (18% over the phone). To evaluate whether the relationship between start channel and churn is significant a Wilcoxon rank sum test is used (as the variable churn is not normally distributed). This test indicates that the null hypothesis is rejected meaning that the mean of churn is significantly different between the two

subgroups of the start channel; online and phone ($W = 34882299$, $p = < 2.2e-16 < 0.05$). More specifically, approximately 52% of the people that categorize as start channel; online have churned, while approximately 34% of the people that categorize as start channel; phone have churned. This allows for the conclusion that people that become a customer through the start channel; online are more likely to churn which is consistent with the initially stated hypothesis in section 2.

5.8 Home label

Figure 3: Histogram Home-label

Since the home label was transformed, the variable has a numerical level of measurement, ranging from 1 to 7, thus the minimum of the home label is one indicating the energy label of the home of the customer is good, while the maximum is seven indicating bad energy label. Conducting the histogram, it is observed that a considerable amount of 4000 home labels are near three which means that the energy label of the home of the customer is relatively good.



Furthermore, it is important to mention that in the amount of 3000 home labels were ranked at 4 and 5, which means the energy label of the home of the customer is neither good nor bad, while in the same frequency a significant amount was ranging as a bad energy label. Subsequently, a logistic regression can be conducted to determine the correlation between home label and churn. The results of the test conclude that home label is significant because the p-value is less than 0.05 ($p < 2e-16$) with a coefficient of 0.234802. This indicates that the relationship between home label and churn is positive which is not consistent with the hypothesis stated in section 2.

6. Modeling

6.1 Baseline model

The baseline model includes the variables: age group, income group, start-channel, and electricity and gas usage as they all show significant relationships with churn when their individual relationships are assessed (previous section). Note that all variables, except start-channel, have a numerical level of measurement. Start-channel is a binary variable and so is the dependent variable churn.

6.2 Logistic regression output

A logistic regression model is useful to use then the dependent variable is binary (churn is a binary variable) and the independent variables are both continuous and binary (start channel is binary and the others are continuous variables)

First model

Regarding the output, the most important is the sign (positive or negative). The output shows that churn decreases when age group and income group increases in value. In other words, the more income people

have and the older they are, then churn will decrease. Regarding the remaining variables, start channel, home label, and electricity and gas usage show that churn will increase when the value of one of these variables will increase. For example, when home-label increases from 1 to 2 (A to B) then it means churn will increase.

More specifically:

- Each one unit increase in age results in a 12.15% decrease in the odds of buying.
- Each one unit increase in the income group results in a 24.55% decrease in the odds of buying.
- The odds are 126% higher of churning when start-channel; online.
- Each one unit increase in home-label results in a 1.1% increase in the odds of buying.
- Each one unit increase in electricity and gas usage results in a 176% decrease in the odds of buying.

Next, the fit criteria. The hit rate is 0.66095 which is above 0.5 indicating the predicted outcome is positive (accurate). The top-decile lift is 1.63028 which is closer to two than one, indicating that it is closer to the model being twice as good in predicting churners (one means the model is not better than random selection). The Gini coefficient is 0.4542665 which is closer to zero than one meaning that the model is not necessarily better than the random selection. However, it is relatively in the middle and the value being closer to one means the model is better, hence it is not sufficient to conclude one or the other. In other words, the model is not fully better than the random selection but also not, not better.

Out of sample validation

The output shows that age group and income group have a negative relationship to churn and the same interpretation as with the first model applies. The specific interpretations are:

- Each one unit increase in age results in a 26.23% decrease in the odds of buying.
- Each one unit increase in the income group results in a 5.67% decrease in the odds of buying.
- The odds are 17.74% higher of churning when start-channel; online.
- Each one unit increase in home-label results in a 0.19% increase in the odds of buying.
- Each one unit increase in electricity and gas usage results in a 24.22% decrease in the odds of buying.

Next, the fit criteria. The hit rate is 0.6626944, the top decile lift is 1.636918, and the Gini coefficient is 0.456452. Since these values are similar to the values of the first model the same interpretations can be applied.

6.3 Stepwise regression

For this model three methods are used, each of their outputs summarized below.

Backward

The final model is according to the backward method is:

$$\text{Churn} \sim 0.3865109 + 0.7772069 * \text{Start_channel} - 0.1568645 * \text{Age_group} - 0.2721492 * \text{Income_group} + 1.0219810 * \text{electr_gas_usage}$$

This model gets a hit rate of 66.15%, indicating the predicted outcome is positive (accurate). The top-decile lift returns a proper value, namely 1.625179 meaning it is better than random selection and the Gini value is mediocre (0.4545637), meaning that the model is not necessarily better than the random selection. In other words, the model is not fully better than the random selection but also not, not better.

The differences in fit criteria when using the out of sample validation model compared to the original model are as follows: the hit rate is higher (66.16%), the top-decile lift is higher (1.632815), and the Gini coefficient is higher (0.4563189). This indicates that the out of sample validation model is better on all three criteria.

Forward

The final model according to the forward method is:

$$\text{Churn} \sim 1.0219810 * \text{electr_gas_usage} - 0.2721492 * \text{Income_group} + 0.7772069 * \text{Start_channel} + 0.1568645 * \text{Age_group} + 0.3865109 * \text{estimation_sample} - 1$$

This model gets a hit rate of 66.1%, indicating the predicted outcome is positive (accurate). The top-decile lift returns a proper value, namely 1.626199 meaning it is better than random selection and the Gini value is mediocre (0.4545637), meaning that the model is not necessarily better than the random selection. In other words, the model is not fully better than the random selection but also not, not better.

The differences in fit criteria when using the out of sample validation model compared to the original model are as follows: the hit rate is lower (65.11%), the top-decile lift is higher (1.632815), and the Gini coefficient is higher (0.4563189). This indicates that the out of sample validation model is better on two out of three criteria.

Both

The final model according to the method using both is:

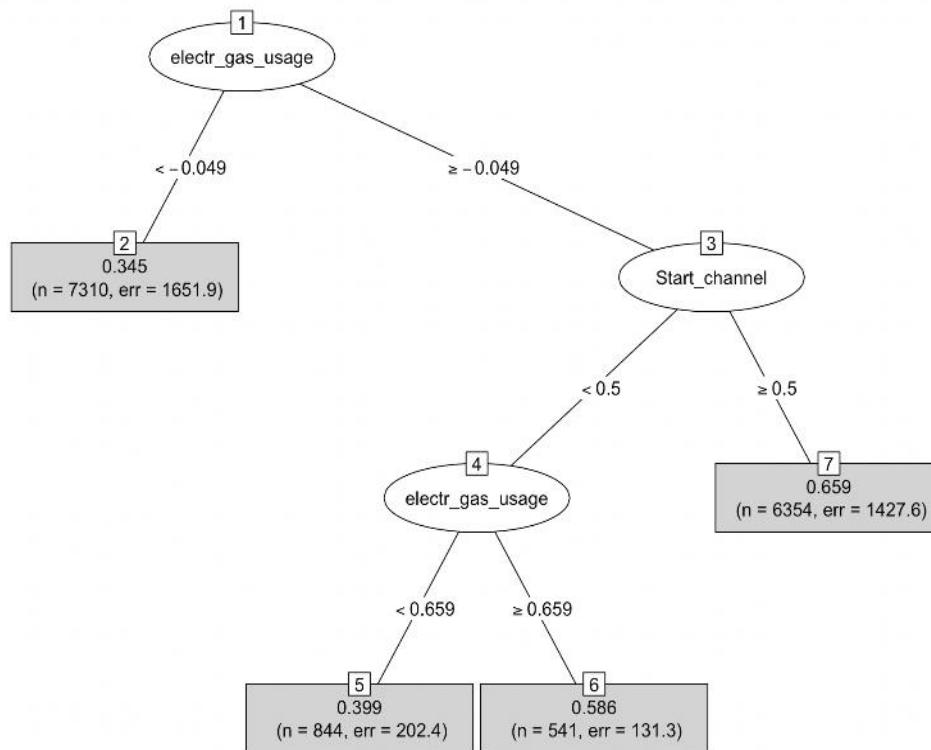
$$\text{Churn} \sim 0.3865109 + 0.7772069 * \text{Start_channel} - 0.1568645 * \text{Age_group} - 0.2721492 * \text{Income_group} + 1.0219810 * \text{electr_gas_usage}$$

What is noticeable is that the backward and both methods give the same final model, indicating that this may be a proper model. Identical to the backward model, this model gets a hit rate of 66.15%, again indicating the predicted outcome is positive (accurate). The top-decile lift returns a proper value, namely 1.625179 meaning it is better than random selection and the Gini value is mediocre (0.4545637), meaning that the model is not necessarily better than the random selection. In other words, the model is not fully better than the random selection but also not, not better.

The differences in fit criteria when using the out of sample validation model compared to the original model are as follows: the hit rate is higher (66.16%), the top-decile lift is higher (1.632815), and the Gini coefficient is higher (0.4563189). This indicates that the out of sample validation model is better on all three criteria.

6.4 Tree model

Figure 4: Tree model



This tree model shows that electricity and gas usage is most important. It also shows that it has an interaction with start-channel, people that use the online channel have a higher churn rate compared to the ones that use the phone. From the people that have start-channel: phone it can be said that the ones that use relatively less than 0.659 have a lower churn than the ones that use more than that.

First model

The hit rate is 66.04% which is above 0.5 indicating the predicted outcome is positive (accurate). The top decile lift is 1.625179 which is above 1 indicating that it is closer to the model being twice as good in predicting churners (1 means the model is not better than random selection). The Gini coefficient is 0.4525951 which is closer to zero than to one meaning that the model is not necessarily better than the random selection. In other words, the model is not fully better than the random selection but also not, not better.

Out of sample validation

The hit rate is 66.29% which is higher than the first model. The top decile lift is 1.567175 is lower than the first model. And the Gini coefficient is 0.4554872 which is higher than the first model. In other words, this model is better for two out of three criteria, but only slightly.

6.5 Random Forest

Figure 5: Random forest plot

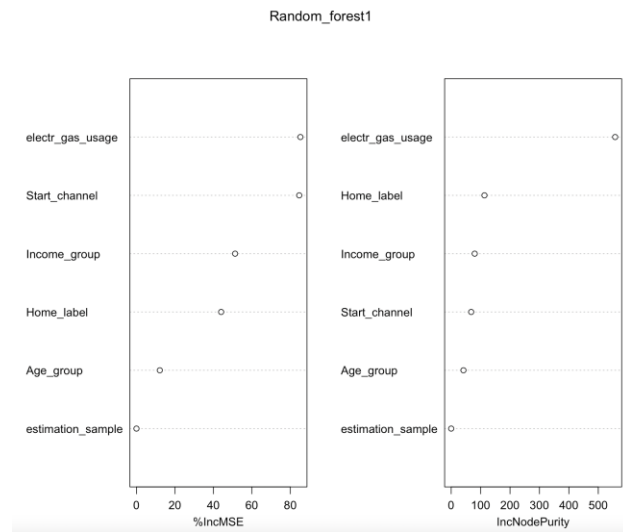


Figure 6: Random forest table

```
> varImpPlot(Random_forest1)
> knitr::kable(importance(Random_forest1))
```

	%IncMSE	IncNodePurity
Age_group	8.712593	31.426569
Income_group	46.667812	66.885489
outlier	6.713385	9.822193
Start_channel	71.635972	62.546494
Home_label	36.051985	96.744625
electr_gas_usage	71.909206	459.128625
estimation_sample	0.000000	0.000000

Diversity is guaranteed by selecting randomly in each split a subset of the variables, estimation sample, age group, home label, income group, start channel and electricity and gas usage. The first column is the mean decrease in accuracy of the predictions when that variable is removed from the model. The second column is a measure of the total decrease in node impurity resulting from splits over that variable (averaged over all of the trees). It is observed that significant decreases occurred in the variables of income group, start channel, home label, and electricity and gas. It is also worth pointing out that in the second column a major decline occurred in electricity and gas usage, approximately 460.

First model

Next, the fit criteria. The hit rate is 0.6731 which is above 0.5 indicating the predicted outcome is positive (accurate). The top decile lift is 1.659865 which is above 1 indicating that it is closer to the model being twice as good in predicting churners (1 means the model is not better than random selection). The Gini coefficient is 0.4831678 which is closer to zero than to one meaning that the model is not necessarily better than the random selection. In other words, the model is not fully better than the random selection but also not, not better.

Out of sample validation

Next, the fit criteria. The hit rate is 0.6647142 which is lower than the first model. The top decile lift is 1.669738 which is higher than the first model. The Gini coefficient is 0.451087 which is lower than the first model. This means that two out of three criteria are lower than the first model, indicating the first model is better.

6.6 SVM

There are four types of support vector machine models, each with different kernels. When using multiple features for support vector machines the preferred kernel is linear, so the following results use a linear kernel.

Figure 7-9: First three (left to right) SVM outputs

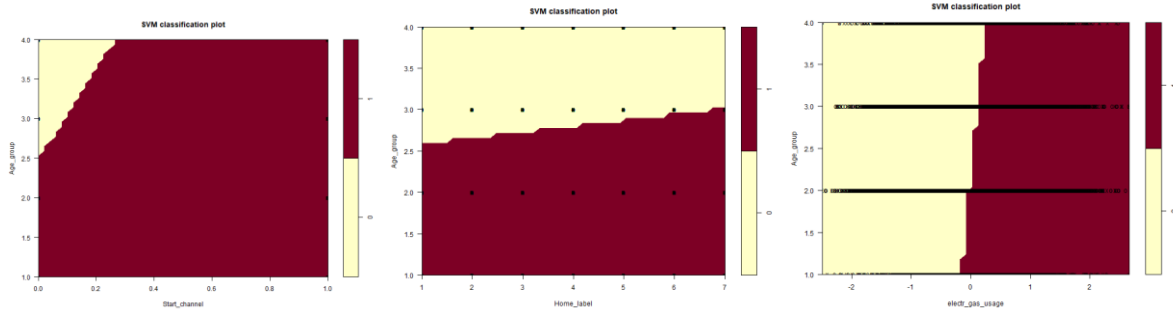


Figure 10-12: Fourth till sixth (left to right) output

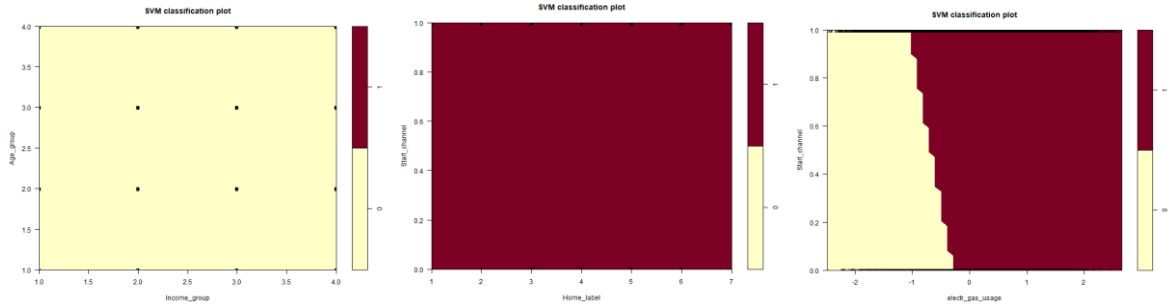


Figure 13-15: seventh till tenth (left to right) SVM output

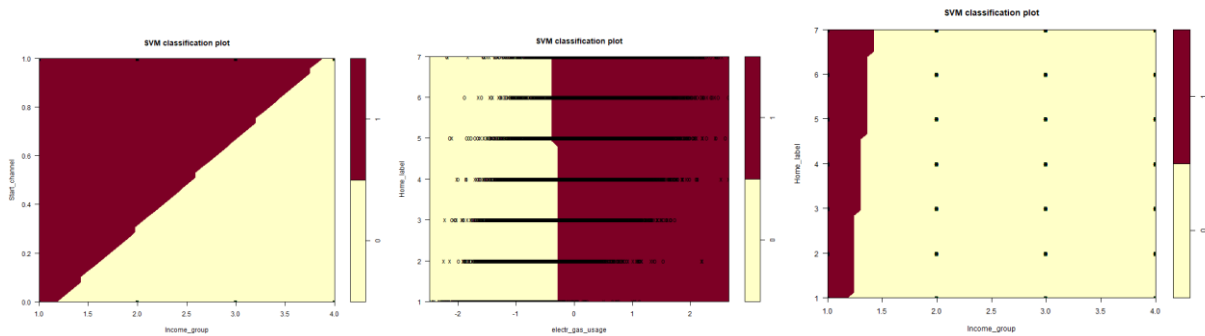


Figure 16: Final SVM output

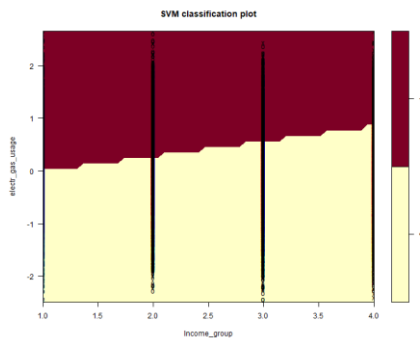


Figure 7: Those that have a start channel which is online are predicted to churn for ages 50 and under.

Figure 8: The worse the energy efficiency the slightly more likely a person is to churn. Ages 50 and under are still predicted to churn with respect to each home label up until the least efficient where the age group 51-67 is predicted to churn.

Figure 9: Those with an electricity and gas usage value of over zero are predicted to churn. With growing age, the slightly higher electricity and gas usage is needed to predict churn.

Figure 10: Nothing of value.

Figure 11: Nothing of value.

Figure 12: Those with a start channel online only need an electricity and gas usage of above -1 to be predicted to churn while others need a value of ca. -0.3.

Figure 13: Those in the lowest income group are all predicted to churn regardless of start channel. The middle-class groups both are predicted to churn when the start channel is online and neither when the start channel is the phone. The highest income group is not predicted to churn regardless of start-channel.

Figure 14: Every home label with an electricity and gas usage of -0.3 and over are predicted to churn. While home labels E, F and G need a slightly lower usage to be predicted to churn.

Figure 15: Only the lowest income group is predicted to churn regardless of energy efficiency.

Figure 16: The lowest income group only needs an electricity and gas usage rate of zero to be predicted to churn. With each step up in the income group the slightly more usage a person needs to be predicted to churn.

Using these predictions, the Support Vector Machine manages to get a hit rate of 66.1%, indicating the predicted outcome is positive (accurate). The top-decile lift returns a proper value of 1.63 and the Gini value is mediocre, 0.45. The out of sample validation gives very similar results to the estimation from the whole dataset so the SVM model is considered a solid option.

7. Managerial Conclusion

This paper explored what factors have an influence on the churn rate with regards to switching energy suppliers. The variables that were considered to be most important in this analysis and accordingly were included in the analysis were: Age (transformed to age group), income (transformed to income group), electricity and gas usage, start channel, and home label. The initial explorative analysis already gave evidence for the hypotheses that were created. As was expected, age and home-label have a negative relationship with churn. Meanwhile, income was predicted to have a positive relationship to churn but it turns out to be negative. This means that if someone increases in age, their income increases, or their home-label becomes worse, they are less likely to churn. The analysis also suggests that, as expected, electricity and gas usage has a positive relationship with churn, meaning that when someone's electricity and gas usage increases, they are more likely to churn. Lastly, it was also found that when someone signs their contract through the website instead of over the phone, they have an 18% higher churn rate. In terms of managerial implications, this shows that the energy supply companies should keep a close eye on the development of their customer base in terms of the previously listed variables. For example, it can be profitable to acquire customers that are of older age and have a higher income as they are less likely to churn, meaning they will remain customers for longer which is what the company aims for. It is also important to be aware of the fact that people that register through the phone are less likely to churn which could be due to the fact that older people are more likely to use this way of communication (compared to the website) which is an indirect explanation of a lower churn rate. Furthermore, companies should consider having a special type of contract for customer's who's electricity and gas usage is relatively high or is expected to increase in the future as these people are more likely to churn. A reason for this could be that they look for energy suppliers that are cheaper than their current one, hence having contracts for such customers that are longer but promise discounts in the future may help retain them as customers. The same

strategy can be applied regarding the variable home-label as it retains customers whose house is expected to get worse in the future.

The model that is recommended for managers to use is the tree model. The tree model is recommended to be used because (1) it is easy to make, (2) contrastingly to the random forest it is easy to interpret as it is a singular tree with one sequence of decisions, (3) differently from the SVM model it works well with large datasets, (4) it allows for both interactions and nonlinearity, and (5) no explicit decisions on which variables to include and structure. To further elaborate on the logistic and stepwise regression models. Both the logistic and stepwise regression models create accurate models according to the hit criteria. However, a major disadvantage for managers that want answers to the research question is that the stepwise regression model does not allow for hypothesis testing and the interpretations need to be done very carefully. And a major limitation of the logistic regression model is that it must assume linear relationships between the independent and the dependent variable, hence not allowing for non-linear relationships to be solved with this model.

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